

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 3 – Low (Pseudo-code Writing)**

Course Code:	CSA0501	Course Title:	Programming in Java
CO Assessed:	CO1 – Precision (BL3,BL4)	Assessment:	Assessment Tool 3 – Pseudo-code Writing
Weightage:		Total Marks:	100
CO1 Statement:	Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)		
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Section Batch:	/ SLOT B	Date:	27.07.2026

Code of Conduct

I M. Saravanan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Saravanan

Instructions

- This test contains 10 questions, each carrying 10 marks. Total: 100 Marks.
- No negative marking. Un answered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

20

20

15

10

15

80
100
88

ANNEXURE

Pseudo - code Writing

1. Student Management System.

START

CLASS Student

PRIVATE name, rollno, marks 1,
marks 2, marks 3

METHOD acceptDetails()

INPUT name, rollno

INPUT marks 1, marks 2, marks 3

END METHOD

METHOD calculateAverage()

$$\text{average} = (\text{marks 1} + \text{marks 2} + \text{marks 3}) / 3$$

RETURN average

END METHOD

METHOD displayResult()

average = calculateAverage()

IF average $\geq$ 50 THEN

PRINT "Pass"

ELSE

PRINT "Fail"

ENDIF

END METHOD

END CLASS

CREATE Student S

CALLS. acceptDetails()

CALL S. displayResult()

STOP

2. Library Book Management

START

CLASS Book

PRIVATE title, author, ISBN, is Available

METHOD SetDetails (t, a, i)

title = t

author = a

ISBN = i

is Available = TRUE

END METHOD

METHOD IssueBook()

IF is Available = TRUE THEN

is Available = FALSE

PRINT "Book Issued"

ELSE

PRINT "Not Available"

ENDIF

END METHOD

METHOD display()

PRINT title, author, ISBN, is Available

CREATE Book b1, b2, b3

CALL b1.display()

CALL b2.display()

CALL b3.display()

STOP

3. Employee Salary Calculation

START

PRIVATE empId, name, basic pay, designation

METHOD accept()

INPUT empId, name, designation, basic pay

END METHOD

METHOD calculate Gross()

HRA = basic pay * 0.20

DA = basic pay * 0.10

gross = basic pay + HRA + DA

RETURN gross

END METHOD

METHOD calculate Net()

gross = calculate Gross()

PF = basic pay * 0.12

tax = gross * 0.05

net = gross - PF - tax

RETURN net

END METHOD

METHOD display pay slip()

PRINT empId, name

PRINT "Gross salary",

calculate Gross()

PRINT "Net Salary",

calculate Net()

END METHOD

ENDCLASS

CREATE Employee e

CALL e.accept()

CALL e.display pay slip()

STOP.

4. Shape Area Calculator using Polymorphism

START

CLASS Shape

METHOD calculateArea()

ENDMETHOD

ENDCLASS

CLASS Circle EXTENDS Shape

radius

METHOD calculateArea()

$area = 3.14 * radius * radius$

PRINT area

END METHOD

END CLASS

CLASS Rectangle EXTENDS Shape

length, breadth

METHOD calculateArea()

$area = length * breadth$

PRINT area

END METHOD

END CLASS

CLASS Triangle EXTENDS Shape

base, height

ENDCLASS

CREATE Circle c

CREATE Rectangle r

CREATE Triangle t

CALL c. calculateArea()

CALL r. calculateArea()

CALL t. calculateArea()

STOP

5. Vehicle Registration System

START

CLASS vehicle

regNo, Owner, fuelType

METHOD displayDetails()

PRINT regNo, Owner, fuelType

END METHOD

END CLASS

CLASS TwoWheeler EXTENDS vehicle

engineCapacity

METHOD displayDetails()

PRINT regNo, Owner, fuelType, engineCapacity

END METHOD

END CLASS

CLASS FourWheeler EXTENDS vehicle

SeatingCapacity, AC

METHOD displayDetails()

PRINT regNo, Owner, fuelType, seatingCapacity, AC

END METHOD

END CLASS

CREATE TwoWheeler t

CREATE FourWheeler f

CALL t.displayDetails()

CALL f.displayDetails()

STOP.

6. Bank Account Hierarchy

START

CLASS BankAccount

accNo, holder, balance

METHOD withdraw(amount)

IF balance $\geq$ amount THEN
balance = balance - amount

ENDIF

ENDMETHOD

ENDCLASS

CLASS Savings Account EXTENDS
Bank Account

METHOD withdraw(amount)

IF balance $\geq$ amount THEN
balance = balance - amount

ELSE
PRINT "Insufficient Balance"

ENDIF

ENDMETHOD

METHOD calculateInterest()

Interest = balance $\times$ 0.05

PRINT interest

ENDMETHOD

ENDCLASS

CLASS Current Account EXTENDS

Bank Account

OverdraftLimit

METHOD withdraw(amount)

IF balance + overdraftLimit >= amount THEN

balance = balance - amount

ELSE

PRINT "Overdraft Limit Exceeded"

ENDIF

ENDMETHOD

ENDCLASS

CREATE SavingAccounts

CREATE CurrentAccountC

CALLS. withdraw(1000)

CALLC. withdraw(5000)

STOP.

7. Hospital Patient Record System.

START

CLASS Patient

PRIVATE Diagnosis, prescription

name, age.

METHOD generateBill()

ENDMETHOD

ENDCLASS

CLASS Inpatient EXTENDS Patient

WardNo, admission date, daily charge

METHOD generate Bill()

bill = daily charge * days

PRINT bill

END METHOD

ENDCLASS

CLASS Outpatient EXTENDS Patient

Appointment Date, Consultation Fee

METHOD generate Bill()

PRINT Consultation Fee

END METHOD

END CLASS

CREATE Inpatient IP

CREATE Outpatient OP

CALL IP.generate Bill()

CALL OP.generate Bill

STOP.

8. E-COMMERCE PRODUCT CATALOG

START

CLASS Product

name, price

METHOD apply Discount()

END METHOD

END CLASS

CLASS Electronics EXTENDS Product

METHOD apply Discount()

Price = Price - (Price * 10/100)

END CLASS

CLASS Clothing EXTENDS Product

METHOD apply Discount()

Price = Price - (Price * 20/100)

END METHOD

END CLASS

CLASS Grocery EXTENDS Product

METHOD apply Discount()

Price = Price - (Price * 5/100)

END METHOD

END CLASS

```
CREATE Electronic e
CREATE Clothing c
CREATE Grocery g
CALL e.apply Discount()
CALL c.apply Discount()
CALL g.apply Discount()
STOP
```

9. University Staff Hierarchy

```
START
CLASS Staff
    PRIVATE id, name
    PROTECTED Salary
    METHOD Calculate Allowance()
        RETURN 1000
    END METHOD
END CLASS

CLASS Teaching staff EXTENDS Staff
    METHOD Calculate Allowance()
        RETURN 2000
    END METHOD
END CLASS
```

10. Smart Home Device Controller.

START

CLASS Smart Device

METHOD execute Command (cmd)

ENDMETHOD

END CLASS

CLASS SmartLight EXTENDS Smart Device

METHOD execute Command (cmd)

If cmd = "On" THEN

PRINT "Light On"

ELSE If cmd = "Off" THEN

PRINT "Light Off"

ELSE

PRINT "Light Off"

ENDIF

ENDMETHOD

END CLASS

CLASS SmartThermostat EXTENDS
Smart Device

METHOD execute Command (cmd)

If cmd = "On" THEN

PRINT "Thermostat Off"

ELSE

PRINT "Temperature"

CLASS Professor EXTENDS Teaching Staff

METHOD Calculate Allowance()

RETURN 5000

END METHOD

END CLASS

CLASS Lecturer EXTENDS Teaching Staff

METHOD Calculate Allowance()

RETURN 5000

END METHOD

END CLASS

CLASS Lecturer EXTENDS Teaching Staff

METHOD Calculate Allowance()

RETURN 3000

END METHOD

END CLASS

CREATE Professor P

CREATE Lecturer L

CREATE Non Teaching Staff n

PRINT P. Calculate Allowance()

PRINT L. Calculate Allowance()

PRINT n. Calculate Allowance()

STOP

ENDIF

END METHOD

ENDCLASS

CLASS SmartLock EXTENDS SmartDevice

PRIVATE PIN

CREATE SmartLight

CREATE SmartThermostat t

CREATE SmartLocks

CALL l. executeCommand ("on")

CALL t. executeCommand ("status")

CALL s. executeCommand ("off")

STOP.